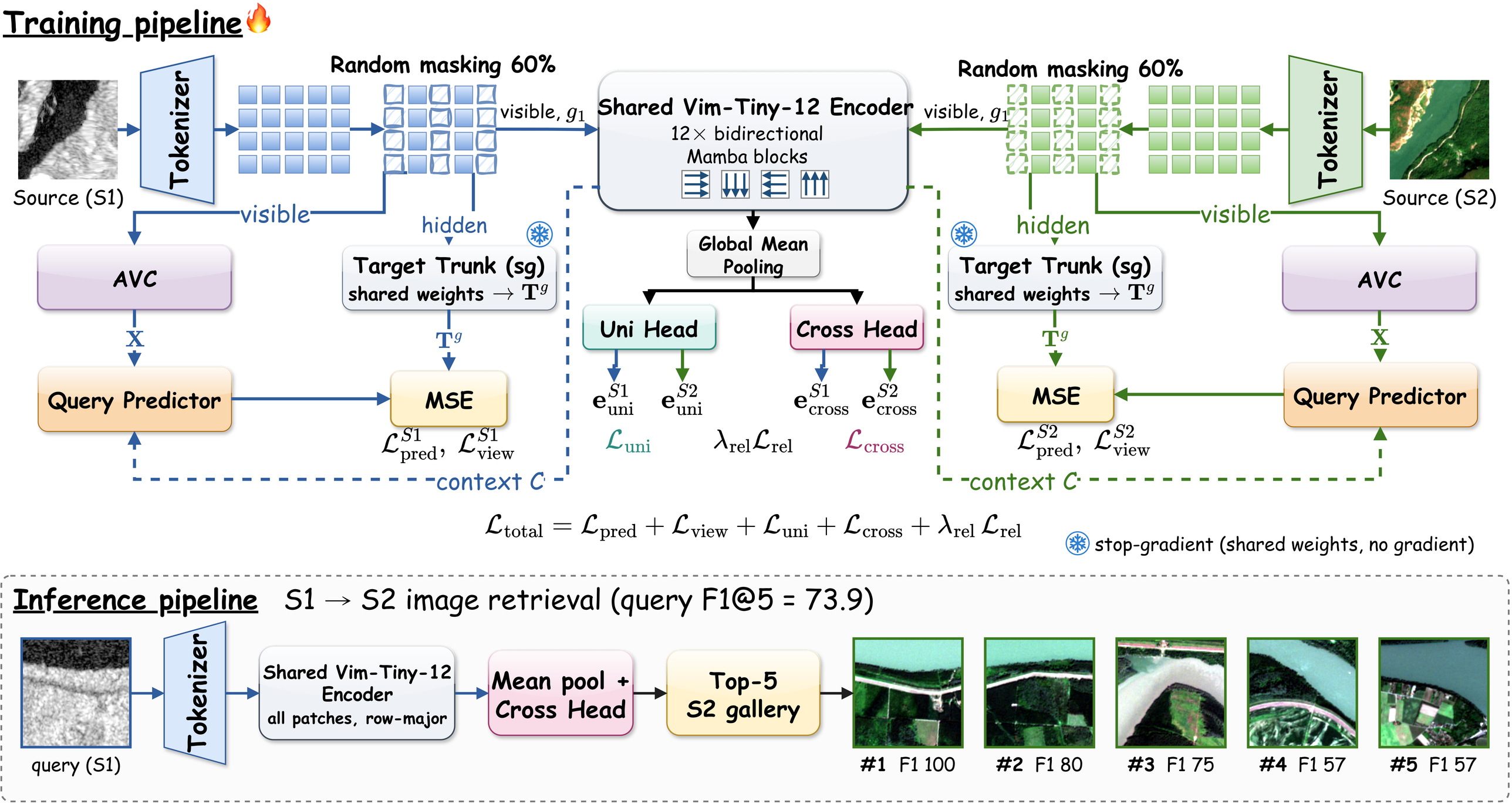

Training pipeline
Source (S1)
Tokenizer
Random masking 60%
visible, g_1
visible
hidden
Shared Vim-Tiny-12 Encoder
12× bidirectional
Mamba blocks
Global Mean Pooling
Uni Head
Cross Head
$\mathbf{e}^{S1}_{\text{uni}}$ $\mathbf{e}^{S2}_{\text{uni}}$
$\mathbf{e}^{S1}_{\text{cross}}$ $\mathbf{e}^{S2}_{\text{cross}}$
$\mathcal{L}_{\text{uni}}$
$\lambda_{\text{rel}}\mathcal{L}_{\text{rel}}$
$\mathcal{L}_{\text{cross}}$
AVC
X
Query Predictor
Target Trunk (sg)
shared weights → $\mathbf{T}^g$
$\mathbf{T}^g$
MSE
$\mathcal{L}^{S1}_{\text{pred}}$, $\mathcal{L}^{S1}_{\text{view}}$
context C
Source (S2)
$\mathcal{L}^{S2}_{\text{pred}}$, $\mathcal{L}^{S2}_{\text{view}}$
$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pred}} + \mathcal{L}_{\text{view}} + \mathcal{L}_{\text{uni}} + \mathcal{L}_{\text{cross}} + \lambda_{\text{rel}}\mathcal{L}_{\text{rel}}$
stop-gradient (shared weights, no gradient)
Inference pipeline
S1 → S2 image retrieval (query F1@5 = 73.9)
query (S1)
Shared Vim-Tiny-12 Encoder
all patches, row-major
Mean pool + Cross Head
Top-5 S2 gallery
#1 F1 100
#2 F1 80
#3 F1 75
#4 F1 57
#5 F1 57